

# Fundamentals Physics

**Tenth Edition**

Halliday



## Chapter 3

Vectors

# 3-1 Vectors and Their Components (1 of 15)

## Learning Objectives

- 3.01** Add vectors by drawing them in head-to-tail arrangements, applying the commutative and associative laws.
- 3.02** Subtract a vector from a second one.
- 3.03** Calculate the components of a vector on a given coordinate system, showing them in a drawing.
- 3.04** Given the components of a vector, draw the vector and determine its magnitude and orientation.
- 3.05** Convert angle measures between degrees and radians.



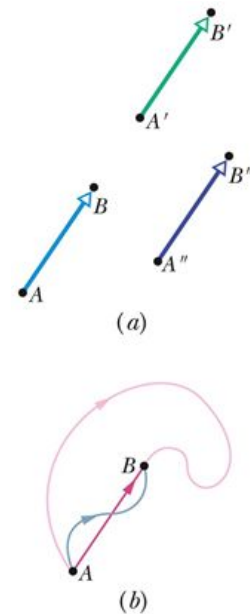
## 3-1 Vectors and Their Components (2 of 15)

- Physics deals with quantities that have both size and direction
- A **vector** is a mathematical object with size and direction
- A **vector quantity** is a quantity that can be represented by a vector
  - Examples: position, velocity, acceleration
  - Vectors have their own rules for manipulation
- A **scalar** is a quantity that does not have a direction
  - Examples: time, temperature, energy, mass
  - Scalars are manipulated with ordinary algebra



# 3-1 Vectors and Their Components (3 of 15)

- The simplest example is a **displacement vector**
- If a particle changes position from  $A$  to  $B$ , we represent this by a vector arrow pointing from  $A$  to  $B$
- In (a) we see that all three arrows have the same magnitude and direction: they are identical displacement vectors.
- In (b) we see that all three paths correspond to the same displacement vector. The vector tells us nothing about the actual path that was taken between  $A$  and  $B$ .



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Figure 3-1

## 3-1 Vectors and Their Components (4 of 15)

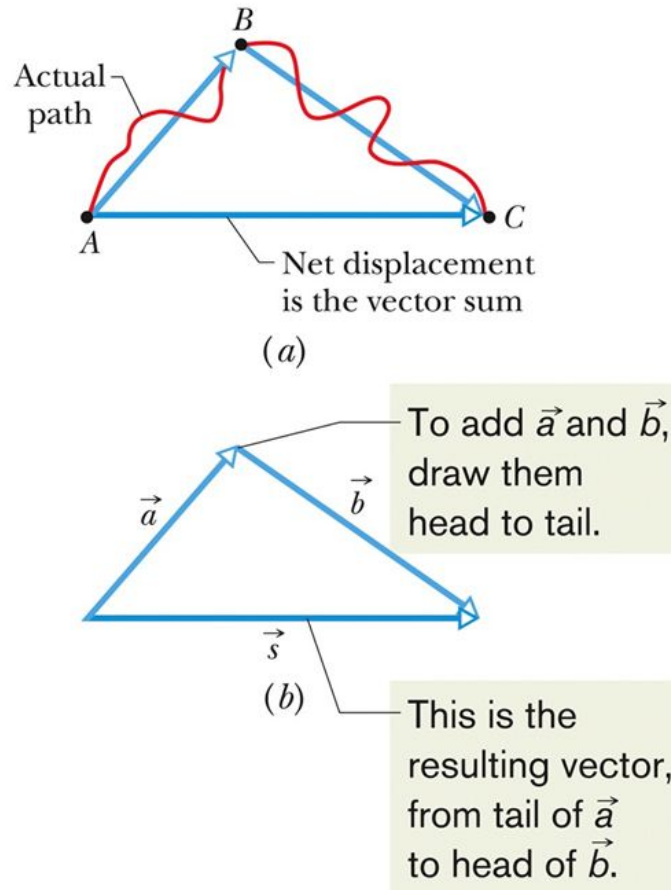
- The **vector sum**, or **resultant**
  - Is the result of performing vector addition
  - Represents the net displacement of two or more displacement vectors

$$\vec{s} = \vec{a} + \vec{b}, \quad \text{Equation (3-1)}$$

- Can be added graphically as shown:



# 3-1 Vectors and Their Components (5 of 15)



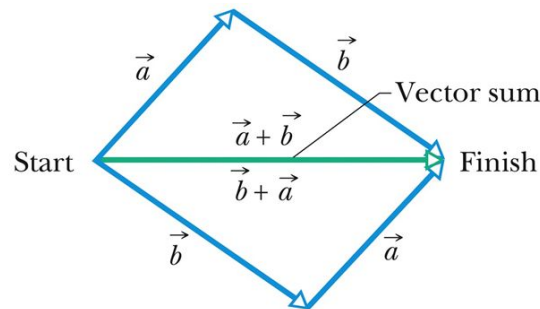
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**Figure 3-2**

## 3-1 Vectors and Their Components (6 of 15)

- Vector addition is **commutative**
  - We can add vectors in any order

$$\vec{a} + \vec{b} = \vec{b} + \vec{a} \quad (\text{commutative law}). \quad \text{Equation (3-2)}$$



You get the same vector result for either order of adding vectors.

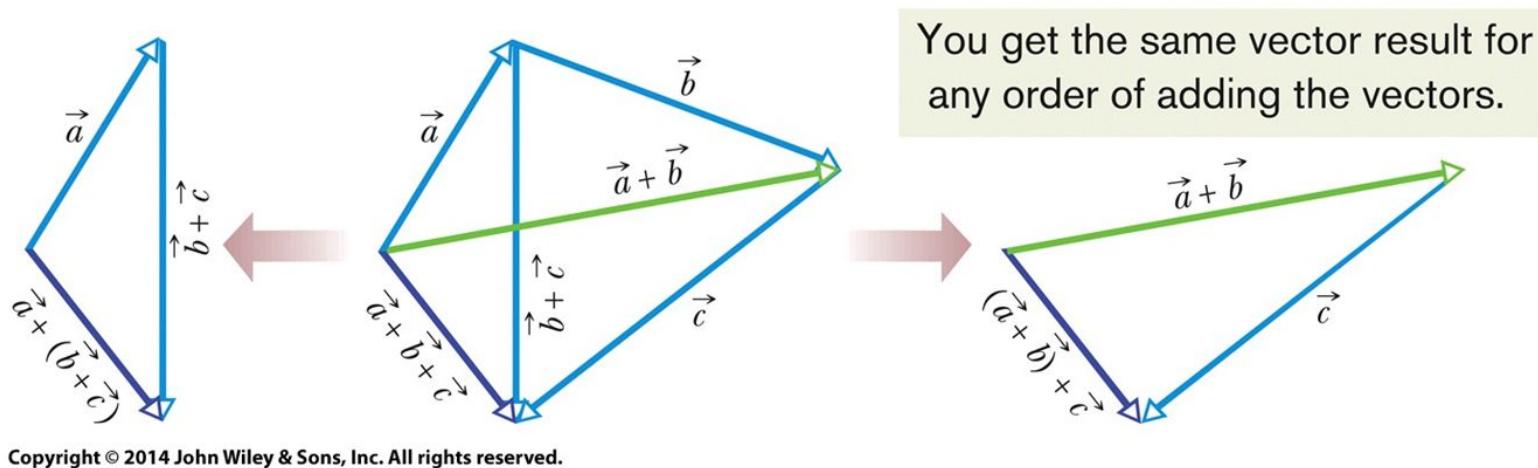
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**Figure (3-3)**

## 3-1 Vectors and Their Components (7 of 15)

- Vector addition is **associative**
  - We can group vector addition however we like

$$\left( \vec{a} + \vec{b} \right) + \vec{c} = \vec{a} + \left( \vec{b} + \vec{c} \right) \quad (\text{associative law}). \quad \text{Equation (3-3)}$$



**Figure (3-4)**



## 3-1 Vectors and Their Components (8 of 15)

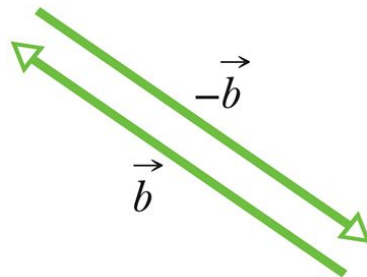
- A negative sign reverses vector direction

$$\vec{b} + (-\vec{b}) = 0.$$

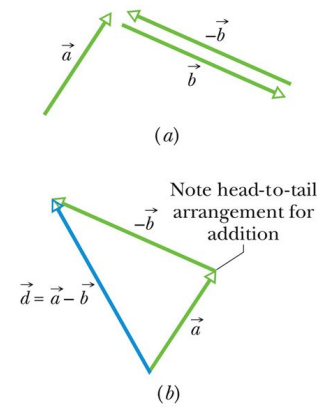
- We use this to define vector subtraction

$$\vec{d} = \vec{a} - \vec{b} = \vec{a} + (-\vec{b})$$

**Equation (3-4)**



**Figure (3-5)**



**Figure (3-6)**

## 3-1 Vectors and Their Components (9 of 15)

- These rules hold for all vectors, whether they represent displacement, velocity, etc.
- Only vectors of the same kind can be added
  - (distance) + (distance) makes sense
  - (distance) + (velocity) does not

## 3-1 Vectors and Their Components (10 of 15)

### Checkpoint 1

The magnitudes of displacements  $\vec{a}$  and  $\vec{b}$  are 3 m and 4 m, respectively, and  $\vec{c} = \vec{a} + \vec{b}$ . Considering various orientations of  $\vec{a}$  and  $\vec{b}$ , what are (a) the maximum possible magnitude for  $\vec{c}$  and (b) the minimum possible magnitude?

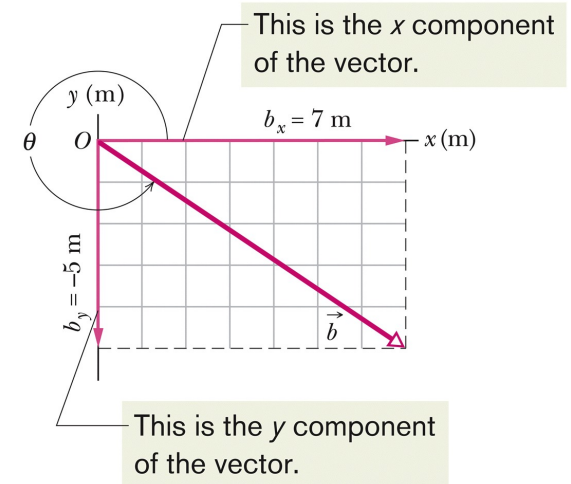
**Answer:**

(a)  $3 \text{ m} + 4 \text{ m} = 7 \text{ m}$

(b)  $4 \text{ m} - 3 \text{ m} = 1 \text{ m}$

## 3-1 Vectors and Their Components (11 of 15)

- Rather than using a graphical method, vectors can be added by **components**
  - A component is the projection of a vector on an axis
- The process of finding components is called **resolving the vector**
- The components of a vector can be positive or negative.
- They are unchanged if the vector is shifted in any direction (but not rotated).



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**Figure (3-8)**

## 3-1 Vectors and Their Components (12 of 15)

- Components in two dimensions can be found by:

$$a_x = a \cos \theta \quad \text{and} \quad a_y = a \sin \theta, \quad \text{Equation (3-5)}$$

- Where  $\theta$  is the angle the vector makes with the positive  $x$  axis, and  $a$  is the vector length
- The length and angle can also be found if the components are known

$$a = \sqrt{a_x^2 + a_y^2} \quad \text{and} \quad \tan \theta = \frac{a_y}{a_x} \quad \text{Equation (3-6)}$$

- Therefore, components fully define a vector

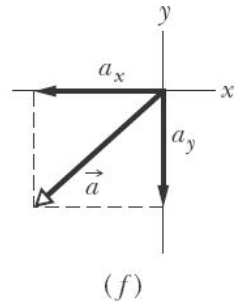
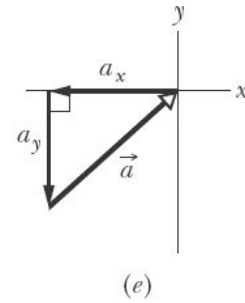
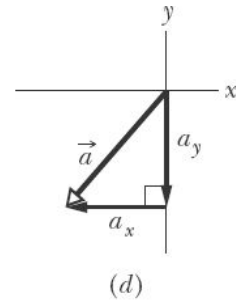
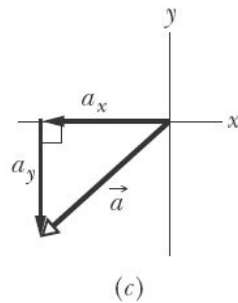
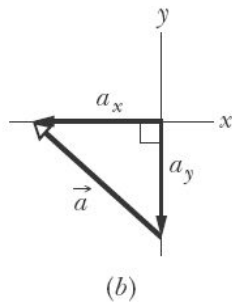
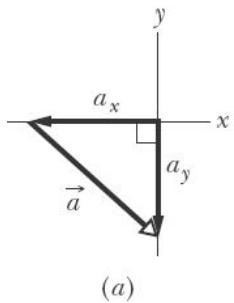
## 3-1 Vectors and Their Components (13 of 15)

- In the three dimensional case we need more components to specify a vector
  - $(a, \theta, \phi)$  or  $(a_x, a_y, a_z)$

## 3-1 Vectors and Their Components (14 of 15)

### Checkpoint 2

In the figure, which of the indicated methods for combining the  $x$  and  $y$  components of vector  $\vec{a}$  are proper to determine that vector?



**Answer:** choices (c), (d), and (f) show the components properly arranged to form the vector

## 3-1 Vectors and Their Components (15 of 15)

- Angles may be measured in degrees or radians
- Recall that a full circle is  $360^\circ$ , or  $2\pi$  rad

$$40^\circ \frac{2\pi \text{ rad}}{360^\circ} = 0.70 \text{ rad.}$$

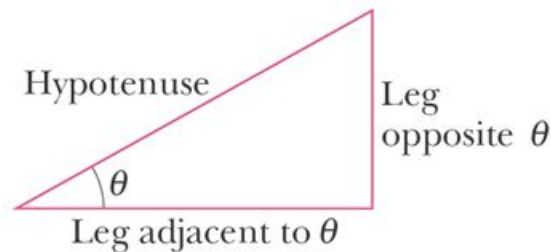
- Know the three basic trigonometric functions

$$\sin \theta = \frac{\text{leg opposite } \theta}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{leg adjacent to } \theta}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{leg opposite } \theta}{\text{leg adjacent to } \theta}$$

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**Figure (3-11)**



## 3-2 Unit Vectors, Adding Vectors by Components (1 of 8)

### Learning Objectives

- 3.06** Convert a vector between magnitude-angle and unit-vector notations.
- 3.07** Add and subtract vectors in magnitude-angle notation and in unit-vector notation.
- 3.08** Identify that, for a given vector, rotating the coordinate system about the origin can change the vector's components, but not the vector itself.

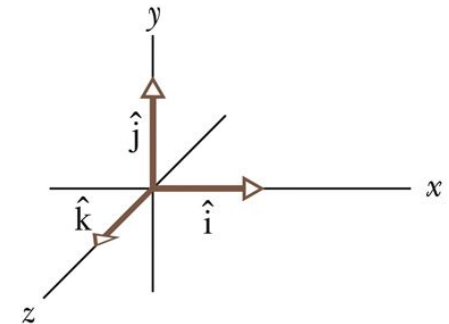
## 3-2 Unit Vectors, Adding Vectors by Components (2 of 8)

- A **unit vector**
  - Has magnitude 1
  - Has a particular direction
  - Lacks both dimension and unit
  - Is labeled with a hat:  $\hat{\phantom{x}}$
- We use a **right-handed coordinate system**
  - Remains right-handed when rotated

$$\vec{a} = a_x \hat{i} + a_y \hat{j} \quad \text{Equation (3-7)}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} \quad \text{Equation (3-8)}$$

The unit vectors point along axes.



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**Figure (3-13)**

## 3-2 Unit Vectors, Adding Vectors by Components (3 of 8)

- The quantities  $a_x \hat{i}$  and  $a_y \hat{j}$  are **vector components**

$$\vec{a} = a_x \hat{i} + a_y \hat{j} \quad \text{Equation (3-7)}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} \quad \text{Equation (3-8)}$$

- The quantities  $a_x$  and  $a_y$  alone are **scalar components**
  - Or just “components” as before

## 3-2 Unit Vectors, Adding Vectors by Components (4 of 8)

- Vectors can be added using components

$$\text{Equation (3-9)} \quad \vec{r} = \vec{a} + \vec{b}, \rightarrow r_x = a_x + b_x$$

$$\text{Equation (3-10)}$$

$$r_y = a_y + b_y$$

$$\text{Equation (3-11)}$$

$$r_z = a_z + b_z.$$

$$\text{Equation (3-12)}$$

## 3-2 Unit Vectors, Adding Vectors by Components (5 of 8)

- To subtract two vectors, we subtract components

$$d_x = a_x - b_x, \quad d_y = a_y - b_y, \quad \text{and} \quad d_z = a_z - b_z,$$

$$\vec{d} = d_x \hat{i} + d_y \hat{j} + d_z \hat{k}. \quad \text{Equation (3-13)}$$

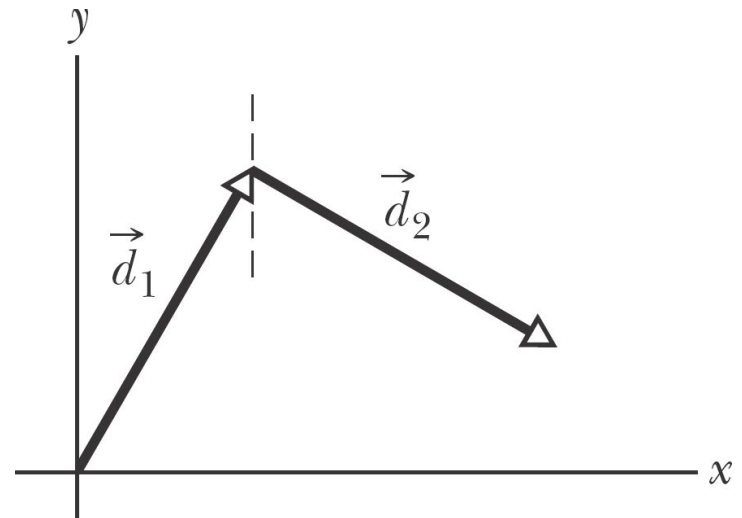
## 3-2 Unit Vectors, Adding Vectors by Components (6 of 8)

### Checkpoint 3

(a) In the figure here, what are the signs of the  $x$  components of  $\vec{d}_1$  and  $\vec{d}_2$ ? (b) What are the signs of the  $y$  components of  $\vec{d}_1$  and  $\vec{d}_2$ ? (c) What are the signs of the  $x$  and  $y$  components of  $\vec{d}_1 + \vec{d}_2$ ?

**Answer:**

- (a) positive, positive
- (b) positive, negative
- (c) positive, positive



## 3-2 Unit Vectors, Adding Vectors by Components (7 of 8)

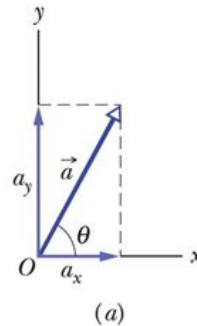
- Vectors are independent of the coordinate system used to measure them
- We can rotate the coordinate system, without rotating the vector, and the vector remains the same

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{a_x'^2 + a_y'^2} \quad \text{Equation (3-14)}$$

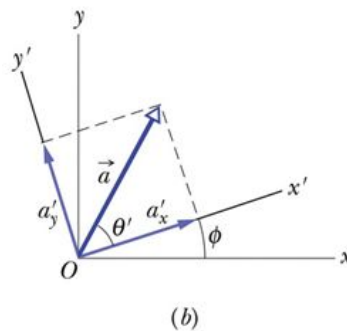
$$\theta = \theta' + \phi. \quad \text{Equation (3-15)}$$

- All such coordinate systems are equally valid

## 3-2 Unit Vectors, Adding Vectors by Components (8 of 8)



Rotating the axes changes the components but not the vector.



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**Figure (3-15)**



## 3-3 Multiplying Vectors (1 of 13)

### Learning Objectives

- 3.09** Multiply vectors by scalars.
- 3.10** Identify that multiplying a vector by a scalar gives a vector, the dot product gives a scalar, and the cross product gives a perpendicular vector.
- 3.11** Find the dot product of two vectors.
- 3.12** Find the angle between two vectors by taking their dot product.
- 3.13** Given two vectors, use the dot product to find out how much of one vector lies along the other.

## 3-3 Multiplying Vectors (2 of 13)

- 3.14** Find the cross product of two vectors.
- 3.15** Use the right-hand rule to find the direction of the resultant vector.
- 3.16** In nested products, start with the innermost product and work outward.

## 3-3 Multiplying Vectors (3 of 13)

- Multiplying a vector  $\mathbf{z}$  by a scalar  $c$ 
  - Results in a new vector
  - Its magnitude is the magnitude of vector  $\mathbf{z}$  times  $|c|$
  - Its direction is the same as vector  $\mathbf{z}$ , or opposite if  $c$  is negative
  - To achieve this, we can simply multiply each of the components of vector  $\mathbf{z}$  by  $c$
- To divide a vector by a scalar we multiply by  $\frac{1}{c}$

## 3-3 Multiplying Vectors (4 of 13)

**Example** Multiply vector  $\mathbf{z}$  by 5

- $\mathbf{z} = -3\mathbf{i} + 5\mathbf{j}$
- $5\mathbf{z} = -15\mathbf{i} + 25\mathbf{j}$

## 3-3 Multiplying Vectors (5 of 13)

- Multiplying two vectors: the **scalar product**
  - Also called the **dot product**
  - Results in a scalar, where  $a$  and  $b$  are magnitudes and  $\phi$  is the angle between the directions of the two vectors:

$$\vec{a} \cdot \vec{b} = ab \cos \phi, \quad \text{Equation (3-20)}$$

- The commutative law applies, and we can do the dot product in component form

$$\vec{a} \cdot \vec{b} = \left( \hat{a}_x \mathbf{i} + \hat{a}_y \mathbf{j} + \hat{a}_z \mathbf{k} \right) \cdot \left( \hat{b}_x \mathbf{i} + \hat{b}_y \mathbf{j} + \hat{b}_z \mathbf{k} \right), \quad \text{Equation (3-22)}$$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}. \quad \vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z. \quad \text{Equation (3-23)}$$

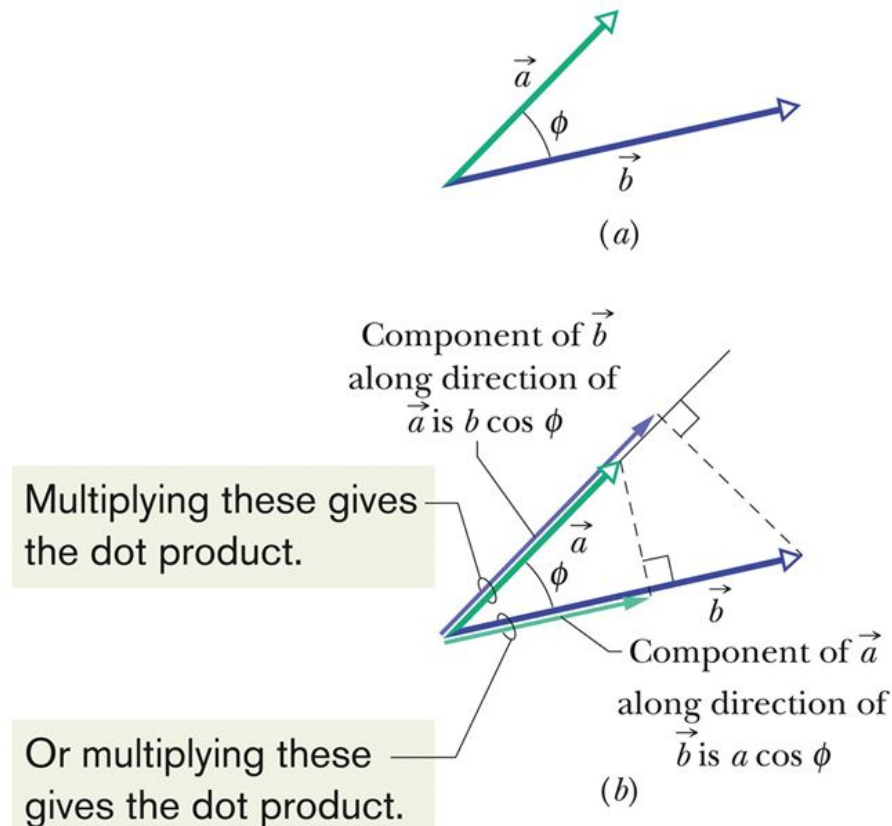
## 3-3 Multiplying Vectors (6 of 13)

- A dot product is: the product of the magnitude of one vector times the scalar component of the other vector in the direction of the first vector

$$\vec{a} \cdot \vec{b} = (a \cos \phi)(b) = (a)(b \cos \phi). \quad \text{Equation (3-21)}$$

- Either projection of one vector onto the other can be used
- To multiply a vector by the projection, multiply the magnitudes

## 3-3 Multiplying Vectors (7 of 13)



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**Figure (3-18)**

## 3-3 Multiplying Vectors (8 of 13)

If the angle  $\phi$  between two vectors is  $0^\circ$ , the component of one vector along the other is maximum, and so also is the dot product of the vectors. If, instead,  $\phi$  is  $90^\circ$ , the component of one vector along the other is zero, and so is the dot product.

### Checkpoint 4

Vectors  $\vec{C}$  and  $\vec{D}$  have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of  $\vec{C}$  and  $\vec{D}$  if  $\vec{C} \cdot \vec{D}$  equals (a) zero, (b) 12 units, and (c)  $-12$  units?

**Answer:**

- (a) 90 degrees
- (b) 0 degrees
- (c) 180 degrees



## 3-3 Multiplying Vectors (9 of 13)

- Multiplying two vectors: the **vector product**
  - The **cross product** of two vectors with magnitudes  $a$  &  $b$ , separated by angle  $\phi$ , produces a vector with magnitude:

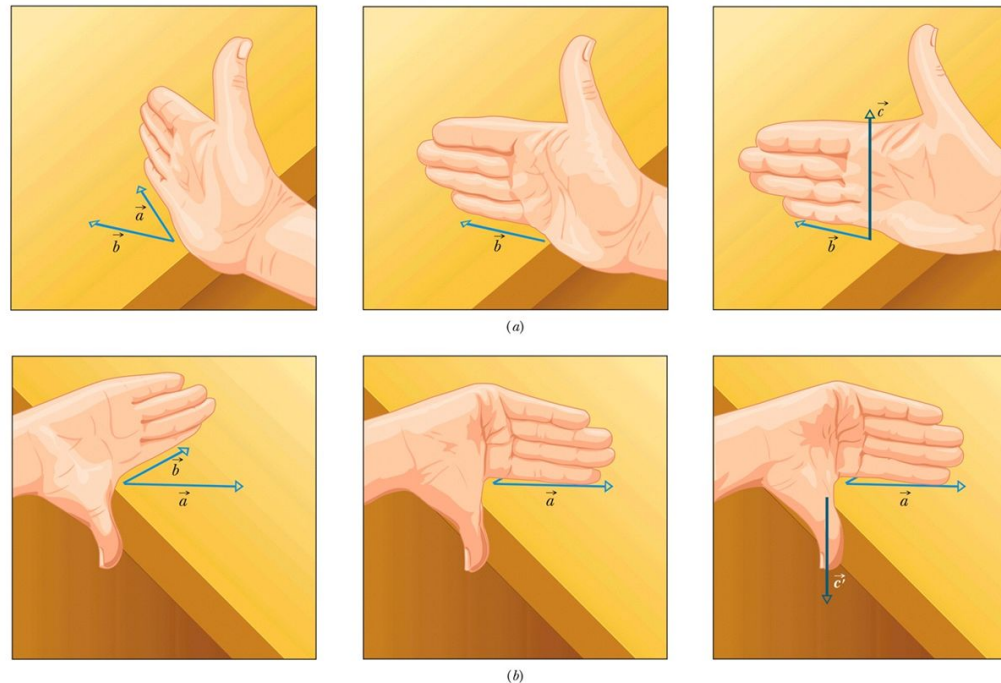
$$c = ab \sin \phi, \quad \text{Equation (3-24)}$$

- And a direction perpendicular to both original vectors
- Direction is determined by the **right-hand rule**
- Place vectors tail-to-tail, sweep fingers from the first to the second, and thumb points in the direction of the resultant vector

## 3-3 Multiplying Vectors (10 of 13)

If  $\vec{a}$  and  $\vec{b}$  are parallel or antiparallel,  $\vec{a} \times \vec{b} = 0$ . The magnitude of  $\vec{a} \times \vec{b}$ , which can be written as  $|\vec{a} \times \vec{b}|$ , is maximum when  $\vec{a}$  and  $\vec{b}$  are perpendicular to each other.

## 3-3 Multiplying Vectors (11 of 13)



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**Figure (3-19)**

The upper shows vector  $a$  cross vector  $b$ , the lower shows vector  $b$  cross vector  $a$

## 3-3 Multiplying Vectors (12 of 13)

- The cross product is not commutative

$$\vec{b} \times \vec{a} = -(\vec{a} \times \vec{b}). \quad \text{Equation (3-25)}$$

- To evaluate, we distribute over components:

$$\vec{a} \times \vec{b} = (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \times (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}), \quad \text{Equation (3-26)}$$

$$a_x \hat{i} \times b_x \hat{i} = a_x b_x (\hat{i} \times \hat{i}) = 0,$$

$$a_x \hat{i} \times b_y \hat{j} = a_x b_y (\hat{i} \times \hat{j}) = a_x b_y \hat{k}.$$

- Therefore, by expanding (3-26):

$$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{i} + (a_z b_x - b_z a_x) \hat{j} + (a_x b_y - b_x a_y) \hat{k}. \quad \text{Equation (3-27)}$$

## 3-3 Multiplying Vectors (13 of 13)

### Checkpoint 5

Vectors  $\vec{C}$  and  $\vec{D}$  have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of  $\vec{C}$  and  $\vec{D}$  if the magnitude of the vector product  $\vec{C} \times \vec{D}$  is (a) zero and (b) 12 units?

Answer:

(a) 0 degrees

(b) 90 degrees

# 3 Summary (1 of 4)

## Scalars and Vectors

- Scalars have magnitude only
- Vectors have magnitude and direction
- Both have units!

## Vector Components

- Given by

$$a_x = a \cos \theta \quad \text{and} \quad a_y = a \sin \theta, \quad \text{Equation (3-5)}$$

- Related back by

$$a = \sqrt{a_x^2 + a_y^2} \quad \text{and} \quad \tan \theta = \frac{a_y}{a_x} \quad \text{Equation (3-6)}$$

## 3 Summary (2 of 4)

### Adding Geometrically

- Obeys commutative and associative laws

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

Equation (3-2)

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c}).$$

Equation (3-3)

### Unit Vector Notation

- We can write vectors in terms of unit vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k},$$

Equation (3-7)

# 3 Summary (3 of 4)

## Adding by Components

- Add component-by-component

$$r_x = a_x + b_x$$

$$r_y = a_y + b_y$$

$$r_z = a_z + b_z.$$

Equations (3-10) - (3-12)

## Scalar Product

- Dot product

$$\vec{a} \cdot \vec{b} = ab \cos \phi,$$

Equation (3-20)

$$\vec{a} \cdot \vec{b} = \left( \hat{a}_x \mathbf{i} + \hat{a}_y \mathbf{j} + \hat{a}_z \mathbf{k} \right) \cdot \left( \hat{b}_x \mathbf{i} + \hat{b}_y \mathbf{j} + \hat{b}_z \mathbf{k} \right),$$

Equation (3-22)



## 3 Summary (4 of 4)

### Scalar Times a Vector

- Product is a new vector
- Magnitude is multiplied by scalar
- Direction is same or opposite

### Cross Product

- Produces a new vector in perpendicular direction
- Direction determined by right-hand rule

$$c = ab \sin \phi, \quad \text{Equation (3-24)}$$

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